M-Embedded Upgrade of Rao's Theorem on Nice Surjections

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Source and Target

The source paper is T. S. S. R. K. Rao, *Nice surjections on spaces of operators*, arXiv:0707.2140. Remark 14 asks whether Rao's preceding theorem remains true if Y is assumed merely to be an M-embedded space with strictly convex dual. The remark identifies the missing point:

> if $\Psi : \mathscr{K}(X,Y) \to \mathscr{K}(X,Y)$ is such that $\Psi(T) = UTV$, then $\Psi^* = \Phi^*|\mathscr{K}(X,Y)^*$. Now $\Psi^{***} = \Phi^*$ on $\mathscr{K}(X,Y)^*$. Since $\mathscr{K}(X,Y)^*$ is weak*-dense in its bi-dual $\mathscr{L}(X,Y)^*$ we get that $\Phi(T) = \Psi^{**}(T) = UTV$.
>
> *Remark* 14. We do not know if the above theorem remains true if one merely assumes that Y is a M-embedded space with a strictly convex dual. It follows from Proposition III.2.2 of [5] that for such a Y any onto isometry is weak*-continuous. To complete the proof as above one would require a into isometry of Y^* to be weak*-continuous.

In text, the source says that to complete the proof one would require an into isometry of Y^* to be $\sigma(Y^*, Y)$-continuous.

Result

We prove that the missing isometry statement is true and then insert it into Rao's proof. Since a nonreflexive M-embedded Y has

$$\mathcal{K}(X,Y)^{**} \cong \mathcal{L}(X,Y^{**}),$$

the natural upgraded version of Rao's theorem is a theorem on $\mathcal{L}(X,Y^{**})$. When Y is reflexive this is exactly Rao's original displayed target space $\mathcal{L}(X,Y)$.

Idea of the Proof

Rao's proof of Theorem 13 first restricts the adjoint of the operator on $\mathcal{K}(X,Y)^{**}$ to $\mathcal{K}(X,Y)^* = X\widehat{\otimes}_\pi Y^*$. Then Rao's Theorem 9 gives a tensor-factor representation

$$\Psi(x \otimes y^*) = Vx \otimes Jy^*,$$

where $V : X \to X$ and $J : Y^* \to Y^*$ are into isometries. In the reflexive case, J is automatically weak-star continuous and hence has a preadjoint $U : Y \to Y$.

For M-embedded Y, this automatic continuity is the only missing point. The decomposition

$$Y^{***} = Y^* \oplus_1 Y^\perp$$

forces every into isometry $J : Y^* \to Y^*$ to preserve the predual Y: if $J^{**}\eta$ had a nonzero Y^*-component for some $\eta \in Y^\perp$, then the ℓ_1-norm identity would force that one vector to be norm-additive with all vectors in $J(Y^*)$, contradicting strict convexity after applying the identity to u and $-u$. Thus J is weak-star continuous, so $J = U^*$. The rest of Rao's proof then gives $\Phi = \psi^{**}$, where $\psi(T) = UTV$.

The Full Upgrade

Theorem 1. *Let X be reflexive and let Y be an M-embedded Banach space. Assume that X and Y^* are strictly convex, that X is not isometric to a subspace of Y^*, and that Y^* is not isometric to a subspace of X^*. Assume also that the canonical identifications used in Rao's proof hold:*

$$\mathcal{K}(X,Y)^* = X\widehat{\otimes}_\pi Y^*, \qquad \mathcal{K}(X,Y)^{**} = \mathcal{L}(X,Y^{**}).$$

Suppose $\Phi : \mathcal{L}(X,Y^{**}) \to \mathcal{L}(X,Y^{**})$ is a bounded linear surjection such that Φ^* maps weak-star-continuous extreme points of the dual unit ball to extreme points.

Then there are $U \in \mathcal{L}(Y)$ and $V \in \mathcal{L}(X)$, with $U^* : Y^* \to Y^*$ and $V : X \to X$ into isometries, such that

$$\Phi(S) = U^{**}SV \qquad (S \in \mathcal{L}(X,Y^{**})).$$

Equivalently, Φ is the bitranspose of the nice surjection

$$\psi : \mathcal{K}(X,Y) \to \mathcal{K}(X,Y), \qquad \psi(T) = UTV.$$

Proof. Put $E = \mathcal{K}(X,Y)$. Under the stated identifications, $E^* = X\widehat{\otimes}_\pi Y^*$ and $E^{**} = \mathcal{L}(X,Y^{**})$. Since Y is M-embedded, Y^* has the Radon–Nikodym property; together with the corresponding hypothesis on X and the metric approximation assumptions used in Rao's proof, this is the setting in which the source identifies E^* with $X\widehat{\otimes}_\pi Y^*$ and uses that the unit ball of E^* is the norm-closed convex hull of its extreme points.

As in Rao's proof of Theorem 13, the hypothesis on weak-star-continuous extreme points implies that

$$\Psi := \Phi^*|_{E^*} : E^* \to E^*$$

is a bounded one-to-one operator preserving extreme points of the unit ball. Rao's Theorem 9, with the same strict convexity and non-isometric-subspace assumptions, gives into isometries $V : X \to X$ and $J : Y^* \to Y^*$ such that

$$\Psi(x \otimes y^*) = Vx \otimes Jy^* \qquad (x \in X,\ y^* \in Y^*).$$

The only place where the reflexivity of Y was used in the original proof is to regard the into isometry J as the adjoint of an operator on Y. We prove that this remains true for M-embedded Y in the lemma below. Therefore there is $U \in \mathcal{L}(Y)$ with $J = U^*$.

Define $\psi : \mathcal{K}(X,Y) \to \mathcal{K}(X,Y)$ by

$$\psi(T) = UTV.$$

For $x \in X$, $y^* \in Y^*$, and $T \in \mathcal{K}(X,Y)$,

$$\langle x \otimes y^*, \psi(T)\rangle = y^*(UTVx) = U^*y^*(TVx) = \langle Vx \otimes U^*y^*, T\rangle.$$

Thus $\psi^* = \Psi$. Since Ψ preserves extreme points, ψ is nice; since Ψ is an into isometry, ψ is a quotient map and hence a nice surjection.

Finally Φ^* and ψ^{***} agree on the canonical copy of E^* in E^{***}. Both maps are weak-star continuous on E^{***} with predual E^{**}, and E^* is weak-star dense in E^{***}. Hence $\Phi^* = \psi^{***}$, so $\Phi = \psi^{**}$. Under $E^{**} = \mathcal{L}(X, Y^{**})$, the bitranspose of $T \mapsto UTV$ is

$$S \mapsto U^{**}SV,$$

which proves the theorem. □

The Missing Lemma

Lemma 1. *Let Y be an M-embedded Banach space and assume that Y^* is strictly convex. Then every linear isometry*

$$J : Y^* \longrightarrow Y^*$$

is $\sigma(Y^*, Y)$-continuous.

Proof. The statement is trivial if $Y^* = \{0\}$, so assume $Y^* \neq \{0\}$. Since Y is M-embedded,

$$Y^{***} = Y^* \oplus_1 Y^\perp,$$

where $Y^\perp = \{\eta \in Y^{***} : \eta(y) = 0 \text{ for all } y \in Y\}$.

Let $J : Y^* \to Y^*$ be a linear isometry. Its bitranspose

$$J^{**} : Y^{***} \to Y^{***}$$

is again a linear isometry, and J^{**} maps the canonical copy of Y^* into itself.

We claim that $J^{**}(Y^\perp) \subseteq Y^\perp$. Fix $\eta \in Y^\perp$, and write

$$J^{**}\eta = a + b, \qquad a \in Y^*, \quad b \in Y^\perp.$$

For every $u \in Y^*$, the ℓ_1-sum norm gives

$$\|u\| + \|\eta\| = \|u + \eta\| = \|J^{**}u + J^{**}\eta\| = \|Ju + a\| + \|b\|.$$

Also

$$\|\eta\| = \|J^{**}\eta\| = \|a\| + \|b\|.$$

Subtracting yields

$$\|Ju + a\| = \|Ju\| + \|a\| \qquad \text{for all } u \in Y^*.$$

If $a \neq 0$, choose $u \in Y^*$ with $Ju \neq 0$. Strict convexity of Y^* forces equality in the triangle inequality only along a single nonnegative ray, so Ju and a must be nonnegative scalar multiples of one another. Applying the same equality to $-u$ forces $-Ju$ and a to be nonnegative scalar multiples of one another. This is impossible. Hence $a = 0$, and $J^{**}\eta \in Y^\perp$.

Thus $Y^\perp$ is invariant under J^{**}. For $y \in Y$ and $\eta \in Y^\perp$,

$$\eta(J^*y) = J^{**}\eta(y) = 0.$$

Therefore $J^*y \in (Y^\perp)_\perp = Y$. Hence $J^*(Y) \subseteq Y$, so J has a preadjoint $J_* : Y \to Y$. Equivalently, $J = (J_*)^*$, and J is $\sigma(Y^*, Y)$-continuous. □

Corollary 1. *Rao's Remark 14 has a positive answer in the natural bidual formulation: the proof of Theorem 13 extends from reflexive Y to M-embedded Y with strictly convex dual after replacing $\mathcal{L}(X, Y)$ by $\mathcal{L}(X, Y^{**}) = \mathcal{K}(X, Y)^{**}$. If Y is reflexive, this is exactly the original target space.*

Proof. The lemma supplies the only missing weak-star continuity step named in Remark 14. The theorem above then repeats Rao's proof of Theorem 13 with that step inserted. □

Verification and Limitations

The proof is formal once the tensor identifications in the theorem statement are available. The packet states these identifications explicitly because the source's approximation-property hypotheses are serving precisely this role. No computational verification is relevant.

The only interpretive caveat is target-space notation: for nonreflexive M-embedded Y, the natural bidual of $\mathcal{K}(X, Y)$ is $\mathcal{L}(X, Y^{**})$, not $\mathcal{L}(X, Y)$. This packet therefore proves the theorem in the natural form suggested by the paragraph immediately preceding Rao's Theorem 13.

Bounded novelty check: run indexes had no full packet for arXiv:0707.2140. Web searches on June 18, 2026 for the exact title, for the exact Remark 14 phrase, and for close phrases such as "M-embedded strictly convex dual into isometry weak-star continuous" did not reveal a later published answer. Local source-index searches likewise found only the original Remark 14 and this run's earlier partial packet.

Human Review Recommendation

Review as a claimed full solution to Rao's Remark 14 in the natural bidual formulation. The highest-value checks are:

1. confirm the tensor identifications under the intended approximation-property hypotheses;
2. confirm that Rao's Theorem 9 proof uses reflexivity of Y only to obtain a preadjoint for the Y^*-isometry;
3. confirm the bitranspose formula $\psi^{**}(S) = U^{**}SV$.

References

[1] T. S. S. R. K. Rao, *Nice surjections on spaces of operators*, arXiv:0707.2140.

[2] P. Harmand, D. Werner, and W. Werner, *M-ideals in Banach spaces and Banach algebras*, Lecture Notes in Mathematics 1547, Springer, 1993.

[3] J. Diestel and J. J. Uhl, *Vector Measures*, Mathematical Surveys 15, American Mathematical Society, 1977.